



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

FEED-OXIDATE EXCHANGER (E-2302A/B)

Docu. No. : 14780-8120-PS-E2302
Project No. : 120117
Rev. No. : Z1
Date : MAY 11, 2016
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TITLE :

FEED-OXIDATE EXCHANGER

(E-2302A/B)

DISCIPLINE : **PROCESS**
DOCUMENT NO. : **14780-8120-PS-E2302**
PROJECT NAME : **PTT Phenol Train II Project**
PROJECT NO. : **120117**
OWNER : **PTT Phenol Company Limited (PPCL)**

Z1	MAY 11, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F2	FEB 11, 2015	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
F1	MAY 28, 2014	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
A1	OCT 8, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
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REV. NO.	DATE	DESCRIPTION	PREP'N	REVIEW	QA REVIEW	APPROVAL
REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW	APPROVAL	
Z1	PROCESS	S.W.Park	J.S.Chung	-	S.W.Lee	
		MAY 11, 2016	MAY 11, 2016	-	MAY 11, 2016	

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SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units

Metric

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Case		Design	TEMA Type (Note 2)	C_U	Orientation	Horizontal	Rev		
Number of Units	1		Shells per Unit	2	Shells in Series	1			
Surface per Unit	m ²	2058.8	Surface per Shell	m ²	1029.4	Shells in Parallel	2		
		(Note 5)			(Note 5)				
PERFORMANCE OF ONE UNIT									
		In	SHELLSIDE		Out	In	TUBESIDE		
Fluid Name				Cumene & CHP				Cumene & CHP	
Total Flowrate (Note 3)		kg/h	189488 × 1.05			1076643 × 1.05			
Vapour		kg/h	0	43393 × 1.05		0	0		
Liquid		kg/h	189488 × 1.05	146095 × 1.05		1076643 × 1.05	1076643 × 1.05		
Fluid Condensed		kg/h	0			0			
Fluid Evaporated		kg/h	43393 × 1.05			0			
LIQUID PROPERTIES									
		SEE PAGES 8, 9				SEE PAGE 7			
Density	kg/m ³								
Viscosity	cP								
Specific Heat	kcal/kg.°C								
Thermal Conductivity	kcal/h.m.°C								
Surface Tension	dyne/cm								
Pour Point Temperature	°C								
VAPOUR PROPERTIES									
		SEE PAGES 8, 9							
Density	kg/m ³								
Viscosity	cP								
Specific Heat	kcal/kg.°C								
Thermal Conductivity	kcal/h.m.°C								
Hydrate Temperature	°C								
THERMAL AND HYDRAULIC DATA									
Operating Temperature	°C	57.7	60.4	83.1	74.6				
Operating Pressure	kg/cm ² g	-0.9478	(Note 4)	3.530					
Number of Passes per Shell		1		2					
Average Velocity	m/s	3.94		1.92					
Pressure Drop (Allow / Calc)	kg/cm ²	0.0643	0.053	0.703	0.487				
Fouling Factor	h.m ² .°C/kcal	0.0004		0.0003					
Duty (Note 3)	kcal/h × 10 ⁶	3.991 × 1.05		MTD	°C	14.0			
U-value Service/Clean	kcal/h.m ² .°C	146	187	Over Surface	%	0.8			
						(Note 6)			
DESIGN CONDITIONS									
		SHELLSIDE		TUBESIDE					
Design pressure max/min	kg/cm ² g	3.5		12.0					
Full Vacuum		Required	at 195 °C	Required	at 83 °C				
Test Pressure	kg/cm ² g	To Code		To Code					
Design Temperature max/min	°C	250	15	120	15				
Corrosion Allowance	mm	TBA		1.5					
Insulation Details		Heat Conservation		Heat Conservation					
Hydrogen Service		No		No					
Sour Service		No		No					
Lethal Service		No		No					
Cyclic Service		No		No					
Cleaning Method		Mechanical		Mechanical					
Stress Relief for Process Reasons		No		No					
Heat Release Curve		Yes		Yes					



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CONSTRUCTION OF ONE SHELL

SHELL	Shell I.D.	mm		Channel I.D. (If different)	mm	1120	Rev
TUBES	Number of Tubes		1362 (Note 10)	Tube Length	mm	5300	
	Tube O.D.	mm	19.05	Tube Thickness	mm	1.651 (Note 9)	
	Tube Gauge	BWG	16	Tube Wall Basis		Average	
	Tube Pitch	mm	25.4	Tube Layout Angle	deg.	90	
	Tube-Tubesheet Joint		Expanded	Tube-Baffle Diam. Clearance	mm	0.38	
BAFFLES	Number of Baffles		10	Baffle Type		Support	
	Baffle Cut	%		Baffle Spacing	mm	520	
	Baffle Cut Orientation		Supports Only	Double Seg. Baffle Overlap	mm		
	Baffle Thick	mm	6.4	Baffle-Shell Diam. Clearance	mm	6.4	
BUNDLE	Impingement Protection		None	Tube Supports		None	
	Pairs of Sealing Strips		0	Sealing Strip Details		None	
	Passlane Sealing		No	Passlane Sealing Details		None	
	U-Bend Diam.	mm	57.15	U-Bend Orientation		Vertical	
	U-Bend Area Included		No	U-Bend Support Details		(Note 7)	
	Longitudinal baffle		No	Longitudinal Baffle Seal		N.A.	

CONNECTIONS

SHELLSIDE						TUBESIDE					
Service	No.	N.B.		Rating	Facing	Service	No.	N.B.		Rating	Facing
						Inlet T1	1	12	Inch	300	RFWN
						Outlet T2	1	12	Inch	300	RFWN
						Vent TV1	1	3/4	Inch	300	RFLWN
						Drain TD1	1	3/4	Inch	300	RFLWN

MATERIALS

Tubes	304L Stainless Steel (Seamless)
Shell / Shell Cover	N.A.
Floating Head Cover	N.A.
Channel / Channel Cover	304L Stainless Steel (UOP Std 3-15)
Tubesheet, Stationary	304L Stainless Steel
Tubesheet, Floating	N.A.
Baffles / Tube Supports	304 Stainless Steel
Nozzles / Nozzle Flange	Shellside: N.A.
Nozzles / Nozzle Flange	Tubeside: 304L Stainless Steel
Bolt and nuts	Per Project Specifications
Gaskets, Shellside	Per Project Specifications
Gaskets, Tubeside	Per Project Specifications
Gaskets, Nozzles	Per Project Specifications

ESTIMATED WEIGHTS

Weights per Shell	Exchanger Weight, Dry	TBA	Exchanger Weight, Wet	TBA
(Tonnes)	Bundle Weight, Dry	5.6		

DESIGN REQUIREMENTS

PED Classification	Shellside- N.A.	Tubeside- N.A.
Project Specifications	UOP 963766 - 401, UOP Std 4-11-7, UOP Std 3-11, UOP Std 3-15, UOP Std 9-12-4, 14780-21A1	
Design Code	ASME VIII Div 1	
Equipment Code	TEMA Class R 9th edition, API 660 8th edition 2007	

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Notes:

- Item has been checked for Flow Induced Vibration. Tube damage is not predicted.
- Exchangers will be installed horizontally, alongside E-2303 into the the Preflash Column V-2301. (Refer to UOP Specification 963766-301 (Vessels)). See pages 6, 10.
Based on detail "A" on page 10, CONTRACTOR to revise the drawing for a TEMA type C front head. OWNER/CONSULTANT approval is required if a TEMA type A or B front head is used.
- Flow and duty include 5% design margin.
- As per UOP 963766-401, the specified inlet pressure is -0.9374 kg/cm²(g). To meet the bundle outlet temperature of 60.4°C, the operating pressure at the bottom of bundle is -0.9478 kg/cm²(g). VENDOR to check and confirm.
- The effective finned area excludes U-bend area and area of tubes at portion of bundle.
- Based on LMTD of 18.1°C for equal cold side temperature of 60.4°C at inlet and outlet, % Over surface is 30.1 %.
- Provide one In bend U-bend support to hold 8 outer most tube rows.
- CONTRACTOR to confirm the required stabbed-in nozzle diameter of 1120 mm and provide vessel flange information/drawing by Exchanger VENDOR to Vessel VENDOR for flanges alignment.
- The following information based on low-fin tubes manufacturer's specification.

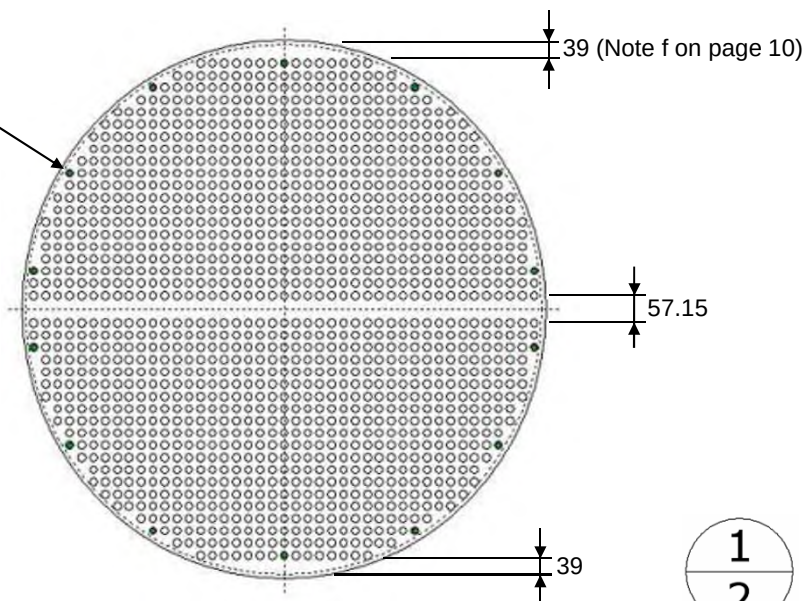
Fin density	1102.4	fin/meter
Plain average tube diameter	19.05	mm
Plain tube thickness (average)	1.651	mm
Fin root diameter	17.272	mm
Fin height	0.889	mm
Fin thickness	0.304	mm
Wall thickness under fins	1.067	mm

Low-fin tubes require no-land (Tube finned in baffles is required).

- For U-tube design, Number of tubes equals number of straights.
- Additional requirements and design notes on page 10 shall be applied.
- Deleted.

Tube Layout

14 tie rods, 12.7 mm dia.
Spacers to suit.



O.T.L. = 1103.9 mm

All dimension in mm
Sketch not to scale

Tubepass Arrangement

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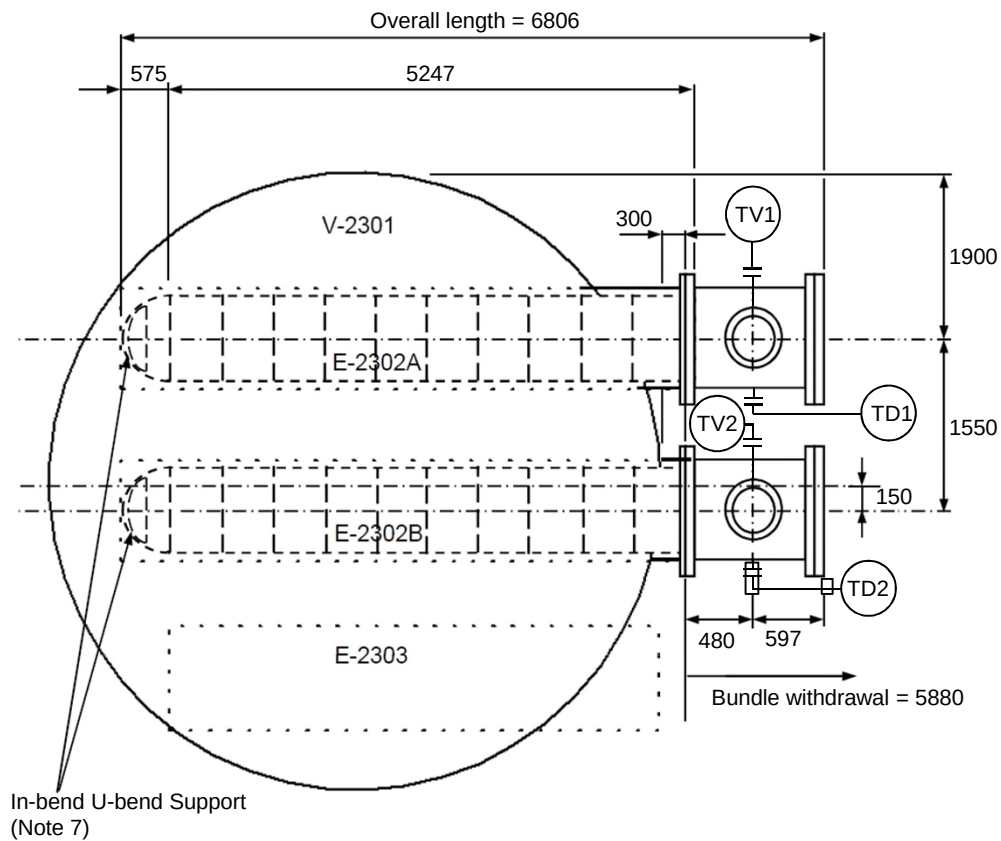
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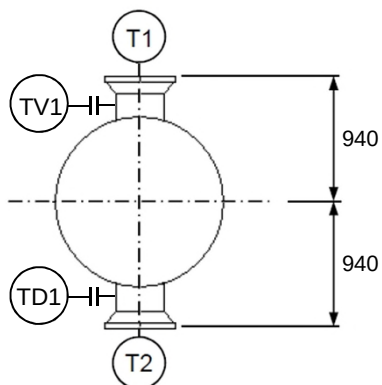
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Setting Plan (Refer to UOP 963766 - 301)



Tubeside Nozzles



Channel ID : 1120 mm

Tube length : 5300 mm

* Deleted.

All dimension in mm
Sketch not to scale

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HotSide Property Curve

Pressure

3.530

kg/cm²g

Temperature °C	Entire Stream Enthalpy kcal/kg	Entire Stream Mass Vapour fraction	Vapour Density kg/m ³	Vapour Viscosity cP	Vapour Therm. Cond. kcal/h.m.°C	Vapour Spec. Heat kcal/kg.°C
83.1	3.70	0.0000	-	-	-	-
82.2	3.29	0.0000	-	-	-	-
81.2	2.88	0.0000	-	-	-	-
80.3	2.47	0.0000	-	-	-	-
79.3	2.06	0.0000	-	-	-	-
78.4	1.65	0.0000	-	-	-	-
77.4	1.23	0.0000	-	-	-	-
76.5	0.82	0.0000	-	-	-	-
75.5	0.41	0.0000	-	-	-	-
74.6	0.00	0.0000	-	-	-	-

Temperature °C	Liquid Density kg/m ³	Liquid Viscosity cP	Liquid Therm. Cond. kcal/h.m.°C	Liquid Spec. Heat kcal/kg.°C	Liquid Surf. Tension dyne/cm	Critical Pressure kg/cm ² [a]	Critical Temperature °C
83.1	841.1	0.577	0.088	0.44	21.4	33.4	353
82.2	842.0	0.583	0.088	0.44	21.5	33.4	353
81.2	842.8	0.590	0.088	0.44	21.6	33.4	353
80.3	843.7	0.596	0.088	0.44	21.7	33.4	353
79.3	844.6	0.603	0.088	0.44	21.8	33.4	353
78.4	845.5	0.610	0.088	0.43	21.9	33.4	353
77.4	846.3	0.616	0.089	0.43	22.0	33.4	353
76.5	847.2	0.623	0.089	0.43	22.1	33.4	353
75.5	848.1	0.631	0.089	0.43	22.2	33.4	353
74.6	849.0	0.638	0.089	0.43	22.3	33.4	353

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ColdSide Property Curve 1

Pressure

-0.8984

kg/cm²g

Temperature °C	Entire Stream Enthalpy kcal/kg	Entire Stream Mass Vapour fraction	Vapour Density kg/m ³	Vapour Viscosity cP	Vapour Therm. Cond. kcal/h.m.°C	Vapour Spec. Heat kcal/kg.°C
55.1	0.00	0.0000	-	-	-	-
61.9	2.74	0.0000	-	-	-	-
68.5	5.47	0.0000	-	-	-	-
75.0	8.21	0.0000	-	-	-	-
81.4	10.95	0.0000	-	-	-	-
87.7	13.69	0.0000	-	-	-	-
93.9	16.42	0.0000	0.5046	0.008	0.0122	0.36
94.5	18.67	0.0242	0.5110	0.008	0.0122	0.36
94.7	20.91	0.0502	0.5110	0.008	0.0122	0.36
95.0	23.15	0.0762	0.5094	0.008	0.0122	0.36

Temperature °C	Liquid Density kg/m ³	Liquid Viscosity cP	Liquid Therm. Cond. kcal/h.m.°C	Liquid Spec. Heat kcal/kg.°C	Liquid Surf. Tension dyne/cm	Critical Pressure kg/cm ² [a]	Critical Temperature °C
55.1	874.2	0.920	0.091	0.40	24.2	33.1	352
61.9	868.1	0.835	0.090	0.41	23.5	33.1	352
68.5	862.1	0.763	0.089	0.42	22.8	33.1	352
75.0	856.1	0.701	0.088	0.42	22.2	33.1	352
81.4	850.1	0.647	0.086	0.43	21.5	33.1	352
87.7	844.2	0.600	0.085	0.44	20.9	33.1	352
93.9	838.3	0.558	0.084	0.45	20.3	33.1	352
94.5	838.8	0.561	0.084	0.45	20.2	33.1	352
94.7	839.6	0.566	0.084	0.44	20.2	33.1	352
95.0	840.6	0.572	0.084	0.44	20.2	33.1	352

ColdSide Property Curve 2

Pressure

-0.9532

kg/cm²g

Temperature °C	Entire Stream Enthalpy kcal/kg	Entire Stream Mass Vapour fraction	Vapour Density kg/m ³	Vapour Viscosity cP	Vapour Therm. Cond. kcal/h.m.°C	Vapour Spec. Heat kcal/kg.°C
55.1	0.00	0.0000	-	-	-	-
61.5	2.61	0.0000	-	-	-	-
67.9	5.21	0.0000	-	-	-	-
74.1	7.82	0.0000	-	-	-	-
80.2	10.42	0.0000	0.3060	0.007	0.0110	0.35
80.8	12.97	0.0273	0.3092	0.007	0.0110	0.35
81.1	15.51	0.0565	0.3092	0.007	0.0110	0.35
81.3	18.06	0.0856	0.3092	0.007	0.0110	0.35
81.6	20.61	0.1147	0.3092	0.007	0.0111	0.35
81.9	23.15	0.1436	0.3092	0.007	0.0111	0.35

Temperature °C	Liquid Density kg/m ³	Liquid Viscosity cP	Liquid Therm. Cond. kcal/h.m.°C	Liquid Spec. Heat kcal/kg.°C	Liquid Surf. Tension dyne/cm	Critical Pressure kg/cm ² [a]	Critical Temperature °C
55.1	874.2	0.920	0.091	0.40	24.2	33.1	352
61.5	868.4	0.839	0.090	0.41	23.5	33.1	352
67.9	862.7	0.769	0.089	0.42	22.9	33.1	352
74.1	857.0	0.709	0.088	0.42	22.3	33.1	352
80.2	851.3	0.657	0.087	0.43	21.7	33.1	352
80.8	851.9	0.661	0.086	0.43	21.6	33.1	352
81.1	852.9	0.669	0.086	0.43	21.6	33.1	352
81.3	854.1	0.678	0.086	0.43	21.5	33.1	352
81.6	855.3	0.687	0.085	0.43	21.5	33.1	352
81.9	856.5	0.698	0.085	0.43	21.4	33.1	351

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ColdSide Property Curve 3

Pressure

-1.0031

kg/cm²g

Temperature °C	Entire Stream Enthalpy kcal/kg	Entire Stream Mass Vapour fraction	Vapour Density kg/m ³	Vapour Viscosity cP	Vapour Therm. Cond. kcal/h.m.°C	Vapour Spec. Heat kcal/kg.°C
55.1	0.00	0.0000	-	-	-	-
55.9	0.31	0.0000	-	-	-	-
56.7	0.62	0.0000	0.1153	0.007	0.0090	0.33
57.4	3.84	0.0341	0.1169	0.007	0.0090	0.33
57.7	7.06	0.0702	0.1169	0.007	0.0090	0.33
58.0	10.28	0.1063	0.1169	0.007	0.0090	0.33
58.3	13.49	0.1422	0.1169	0.007	0.0090	0.33
58.7	16.71	0.1780	0.1169	0.007	0.0091	0.33
59.1	19.93	0.2136	0.1169	0.007	0.0091	0.33
59.6	23.15	0.2490	0.1169	0.007	0.0092	0.33

Temperature °C	Liquid Density kg/m ³	Liquid Viscosity cP	Liquid Therm. Cond. kcal/h.m.°C	Liquid Spec. Heat kcal/kg.°C	Liquid Surf. Tension dyne/cm	Critical Pressure kg/cm ² [a]	Critical Temperature °C
55.1	874.2	0.920	0.091	0.40	24.2	33.1	352
55.9	873.5	0.910	0.091	0.40	24.1	33.1	352
56.7	872.8	0.899	0.091	0.40	24.0	33.1	352
57.4	873.7	0.909	0.090	0.40	24.0	33.1	352
57.7	875.2	0.928	0.090	0.40	23.9	33.1	352
58.0	876.8	0.948	0.089	0.40	23.9	33.1	352
58.3	878.6	0.970	0.089	0.40	23.8	33.1	351
58.7	880.4	0.994	0.089	0.40	23.8	33.1	351
59.1	882.5	1.021	0.088	0.40	23.7	33.2	351
59.6	884.7	1.051	0.088	0.40	23.6	33.2	350

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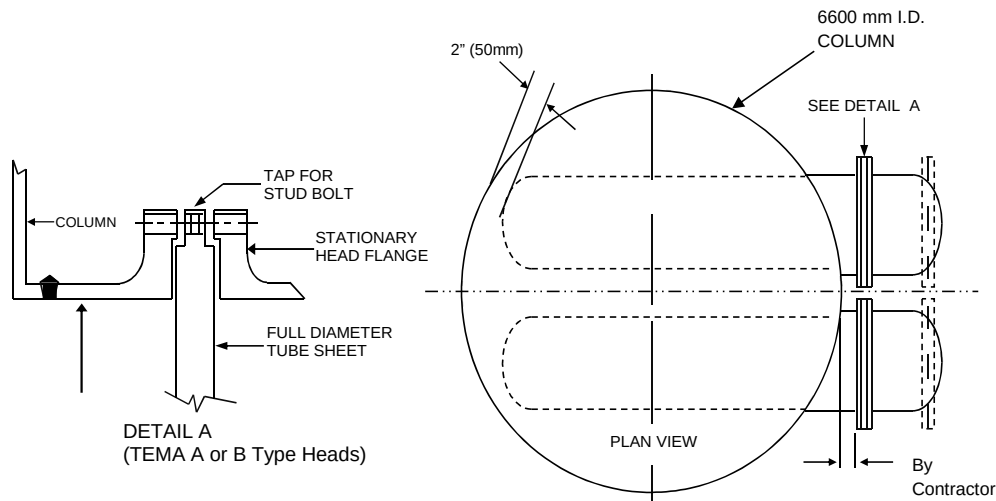
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DUAL STABBED-IN REBOILER REQUIREMENTS (Note 11)



1. REQUIREMENTS

- Bundle vendor shall provide the necessary welding neck flange to column fabricator for attachment to column nozzle.
- Responsibility for design of detail "A" closure is fully chargeable to tube bundle designer except it shall be the TEMA "R" confined gasket type.
- Bundle vendor shall include stationary head together with necessary bolts, nuts and gaskets for securing bundle to column nozzle.

2. TUBE BUNDLE DESIGN:

- Tube bundle shall be U-tube type complying with above dimensions and with a nominal tube length that assures full traversing of column diameter per the dotted rectangle. U-tubes shall conform to UOP Standard Specification 4-11, Section 3.1. For HF Alkylation process applications where process stream contains acid, U-bends shall be stress relieved after bending per Section 4.4.a.12.
- Tube joints shall be the TEMA "R" double groove, rolled type. When welding of tube ends to tubesheet is required it will be noted on project item specification sheet.
- Occasionally this specification sheet will indicate a dimension from column wall to face of nozzle flange in excess of 12" (300 mm). This procedure is used for procuring an increase of bundle tube length at the expense of the neck extension versus the added cost of swaging the column size to a larger diameter. This, however, can be effective only when the tube bundle diameter is limited so there is an annulus in the connection neck that permits the liquid to flow in and the vapor to flow out of this portion of the bundle in an orderly manner.
- Bolt holes in tube sheet shall be drilled and tapped to form joint that permits removal of head while column connection to tube sheet joint remains tight. Bolt holes shall straddle vertical centerline.
- Column fabricator will provide two slide rail supports, for each bundle, installed inside column at 30° off bottom vertical centerline of bundle nozzle. Slide faces will be flush with surface of nozzle. Bundle support plate(s) shall be designed to slide on above rails.
- Provide a minimum of 1" (25 mm) clearance, above and below tube bundle, to permit flow into and out of portion of bundle that projects out from column.
- Design the tube bundle for vibration per UOP Standard Specification 4-11, Section 2.4.
- When a TEMA "C" type front head is specified, the bolting of the channel and integral tubesheet to the column nozzle flange shall be typical of a TEMA "C" type head to a shell.

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